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The need for policy coherence to trigger a transition to biogas production

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ABSTRACT

Transitions towards more sustainable societies involve policy changes cutting across multiple sectors. Ideally, policies targeting different sectors create a coherent push for the adoption of more sustainable solutions. Sustainability transition studies have, however, paid little attention to the role of policy interaction across different policy domains. By focusing on biogas production in Finland and by further developing the technological innovation systems functions in connection to policy coherence, this paper examines how policy coherence is related to triggering transition from the perspective of biogas actors. The results demonstrate how supportive policies in one sector are made inefficient by unsupportive policies, instruments and practices in others. However, the lack of policy coherence, especially at the local level, may also have innovation-triggering influence when it forces actors to consider unconventional solutions. Thus the innovation effects of policy coherence are difficult to foresee and require an actor-focused perspective for analysis.

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1. Introduction

Recently it has been acknowledged that transitions towards more sustainable societies involve policy changes cutting across multiple sectors and taking on a more systemic perspective (Markard

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et al., 2012; Weber and Rohracher, 2012; Kivimaa and Virkamäki, 2014). This systemic perspective links to the concept of 'policy coherence' that has recently evolved as a key term in the analysis of policy mixes and policy interaction in the context of environmental change (Nilsson et al., 2012). Simultaneously, the role of actors in transitions has received increasing focus in transitions literature (e.g. Shove and Walker, 2010; Wirth et al., 2013), while it is a missing perspective in policy coherence studies.

New policies are rarely designed in a vacuum but, rather, layered on pre-existing policies, making changes, introducing new instruments and aligning instruments to better fit to new goals (Kern and Howlett, 2009). Even when focused on a single sector, this layering of policies contains a high risk for incoherent policy mixes, if the whole policy context is not fully reviewed. These policy mixes can be inefficient in triggering transitions, for example due to old instruments undermining new goals, or the prevalence of old policy discourses (Kern and Howlett, 2009; Smith et al., 2010; Kern, 2011). When the perspective is widened to comprise cross-sectoral matters, there is a strong likelihood of the emerging coherence issues being even wider, for example by making rebound effects visible (see van den Bergh, 2013). The policy coordination failure has been recognised as a central problem in governing for innovations in sustainability transitions (Weber and Rohracher, 2012). Not all policies can be coherent, nor is it desirable due to competing interests (Jordan and Halpin, 2006; Carbone, 2008). Yet the key question is how to recognise and avoid unnecessary coherence problems to efficiently promote sustainability. This paper aims to initiate discussion on how policy coherence influences sustainability transitions and create an initial understanding of it by examining a particular case: the adoption of biogas technologies in Finland.

Here, policy coherence is understood as the consistency of expressed policy goals, instruments and other policy-related signals from the perspective of actors affected by the policies (May et al., 2006; Mickwitz et al., 2009) including implementation. Thus, actors are at the centre of policy coherence analysis, indicating the role of policy coherence in inhibiting or promoting sustainability transition. To assess the influence of policy coherence on biogas innovation, we adopt the Technological Innovation Systems (TIS) functions, similarly to Kivimaa and Virkamäki (2014), but extend them by cross-sectoral and actor-based dimensions.

Biogas production in anaerobic co-digestion systems is a boundary-crossing innovation (cf. Raven and Verbong, 2009) between several socio-technical regimes: energy production, waste management, agriculture and transport being the most obvious ones (see e.g. Lantz et al., 2007; Negro et al., 2007; Wilkinson, 2011; Lybaek et al., 2012). Innovations crossing the boundaries of multiple regimes are particularly fruitful for transition experiments (Raven and Verbong, 2009). Furthermore, biogas production deals with both system improvements and system innovations that have been identified as the key points in addressing lock-in (Kemp and Rotmans, 2005). From the perspective of biogas systems, policies targeting multiple sectors have been important in guiding the development of biogas technologies (Negro et al., 2007; Wilkinson, 2011; Lybaek et al., 2012). However, the role of policy coherence for biogas innovation has not previously been studied. Therefore, this paper examines: (1) how local actors experience policy coherence related to biogas production; and (2) how their experiences on policy coherence influence innovation processes that potentially lead to wider transition.

The paper is structured as follows: next, the sustainability transitions approach via Technological Innovation Systems and the analysis of policy coherence are contextualised. Section 3 presents the case study and methods. The results are presented in Sections 4 and 5, followed by discussion and conclusions in Sections 6 and 7.

2. Policy coherence and sustainability transitions

2.1. Policy coherence

Broadly, policy coherence means that policy goals, instruments and other policy-related signals (for example, communication on future changes to policies) are consistent with each other (May et al., 2006; Mickwitz et al., 2009) and that this compatibility occurs across substantive policy domains (e.g.

energy, agriculture) (Nilsson et al., 2012). Ideally, this leads to an increasing likelihood that policies will reach their intended outcomes, such as more sustainable societal change. As an analytical concept, policy coherence is ambiguous and lacks a fixed definition (Rogge and Reichardt, 2013). It is also closely related to several other terms, such as policy integration, policy interaction, policy interplay and policy mixes, all of which deal with the compatibility of policies.

In previous studies, policy coherence has mainly been analysed from the perspective of environmental and climate policies (e.g. Mickwitz et al., 2009; Nilsson et al., 2012) or within the wider context of EU policy (e.g. Carbone, 2008; Hommels et al., 2013). Common to all the assessments is the examination of the interplay of policies across policy domains and across different layers of policies from policy goals to instruments and implementation.

Compared to policy coherence, policy integration deals with the incorporation of specific policy goals and processes into a wider set of policy domains (typically involving environmental policy goals) (Kivimaa and Mickwitz, 2006; Jordan and Lenschow, 2010), potentially facilitating policy coherence. Policy interplay, in turn, focuses on the interactions of different policies (Urwin and Jordan, 2008) and policy mixes deal with different policy measures influencing the same target (Howlett and Rayner, 2007), such as innovation (Flanagan et al., 2011).[1] Policy coherence is a broader concept partially encompassing the aforementioned concepts. However, it is more specific than the above regarding the actualisation of policies at the stakeholder level (Hommels et al., 2013). This means that the policy coherence analysis involves policy implementation and outcomes as actor decisions, going further than the other concepts. It also means that policy coherence involves a temporal perspective as both current state and the evolvement of policies are important (Carbone, 2008) for the consistency of policy signals.

Policy coherence has previously been analysed top-down, searching for mismatches and synergies in policy goals and instruments within and between different sectors (Nilsson et al., 2012; Kivimaa and Virkamäki, 2014). Although beneficial, this kind of analysis faces two complications: first, the identification of the relevant policies and, second, the measuring of policy coherence from often vaguely stated policy goals or preambles to statutes (May et al., 2006). These complications can be handled via utilising a more bottom-up perspective on policy coherence, namely by focusing on local and regional actors' perceptions. These actors provide an understanding of which policies are relevant and related to their activities. They also know what the effects of the policies in practice are. Furthermore, the bottom level actors will provide a perspective throughout the policy process, from goal setting to outcomes, pointing out also the policy-related reasons for the particular outcomes. For example, policies identified as being incoherent by looking at policy outputs may actually appear coherent to grassroots level actors, if civil servants implement them in a coherent manner. For example, forestry officials in Finland combined the unconnected aims of rural, forestry and energy policies to form coherent bioenergy projects (Leskinen, 2006).

Combining these insights and building on Nilsson et al. (2012), we analyse policy coherence via three dimensions: (1) internal, (2) external, and (3) temporal (Fig. 1). Internal policy coherence focuses on one policy domain. The different policy goals, instruments and implementation are assessed horizontally (e.g. by comparing different goals) and vertically (e.g. by comparing goals with instruments) to determine whether they are in line with one another (Nilsson et al., 2012). Policy outcomes are assessed in relation to policy goals. External policy coherence addresses consistency between policy domains.[2] This again involves goals, instruments and implementation but examines the compatibility of, e.g. goals from one policy domain against the instruments in another policy domain. Temporal policy coherence means the consistency and predictability of policies over time. The analysis of policy coherence aims to point out where policies give inconsistent signals to actors, leading to different interpretations on what kind of action is promoted.

[1] Although Rogge and Reichardt (2013) have also presented a broader conceptualisation around the policy mix concept. Their understanding of policy mix comes very close to our policy coherence.

[2] This relates to socio-technical regime interaction (e.g. Raven and Verbong, 2009), while policy domains are often narrower than regimes.

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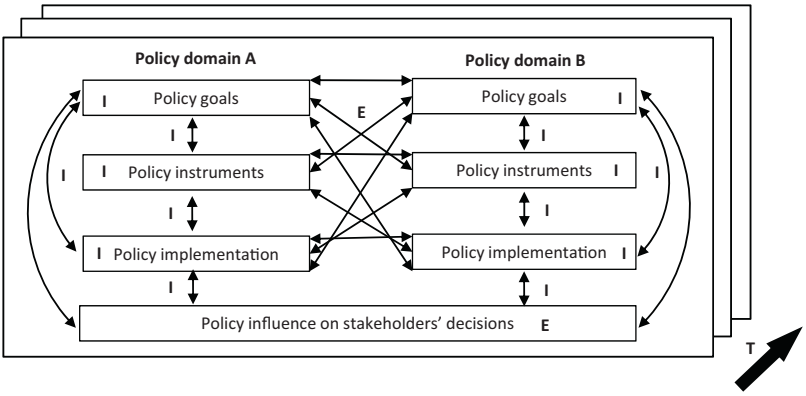

Fig. 1. Dimensions of policy coherence (modified from Nilsson et al., 2012). Internal coherence is marked with I, external coherence with E and temporal coherence with T.

2.2. Policies and innovations

Literature addressing the effects of policy instruments on eco-innovations is extensive and dates all the way back to the 1980s (see a full review by Kemp and Pontoglio, 2011). The recent literature dealing with the innovation effects of environmental policies suggests that the mix of policy approaches and instruments in place in given contexts is relevant from the perspective of innovation, while it is by no means the only important criterion (other examples being stringency, predictability and timing of policies; Kivimaa, 2008, Kemp and Pontoglio, 2011; Rogge and Reichardt, 2013). In particular, the predictability of policies (e.g. Norberg-Bohm, 1999) relates to the temporal dimension of policy coherence. Acknowledging that policy coherence is not the only factor of importance for innovation, this paper focuses on the combined impacts of multiple policies on biogas innovation.

The innovation impacts are explored by adopting and extending the functions of the Technological Innovation Systems (TIS) approach, developed to analyse factors affecting innovation processes (e.g. Suurs et al., 2010). The functions outline the elements needed in innovation processes aiming for more systemic change. This paper adapts an analytical framework first used by Kivimaa and Virkamäki (2014) on policy analysis but adopts a differing angle. Rather than examining policies influencing multiple types of innovation in one sector (transport), we set out to example innovation in one technological field (biogas) and the combined impact of policies from four different domains. This setting also develops a new application of TIS that is generally used in studying the development of a particular technological field.

The TIS functions have not typically been used for policy analysis but were chosen, because TIS is one of the favoured approaches in sustainability transitions research; the intention of its analyses has been to inform policymaking (Markard et al., 2012); and it provides a suitably detailed categorisation of innovation-triggering functions that policies can explicitly or implicitly support or hinder (Kivimaa and Virkamäki, 2014). In fact, many TIS functions link to the elements that Porter and van der Linde (1995) identified as benefits through which environmental policies may lead to innovation.

The TIS framework helps to understand the process of development, diffusion and use of a particular technology by examining components that influence the innovation process for that technology (Suurs and Hekkert, 2009). The innovation process is conceptualised in system functions that are important for innovation triggering and spreading (Table 1). The system functions aid in mapping policies, identifying sources of policy failure and provide justification for policy intervention (Bergek et al., 2008) – an element which is lacking from transition management approaches (Weber and Rohracher, 2012). Policy instruments can be designed to influence one or several critical system functions but political intervention related to all the system functions is not targeted (Kivimaa and Virkamäki, 2014).

Table 1
Analytical framework for using technological innovation systems for policy analysis (slightly modified from Kivimaa and Virkamäki, 2014).

System function	Description	Type of policies related to the function
Knowledge development and diffusion	Knowledge base and networks; how the knowledge is developed, combined and diffused	R&D funding schemes, educational policies, informational instruments
Influence on the direction of search	Incentives and pressures for organisations to enter a field, including visions, expectations and demand	Policy discourse, targets set in strategies, regulations, tax incentives, foresight exercises
Entrepreneurial experimentation	Testing of new technologies, applications and markets in a way that new entrepreneurship and diversification forms	Funding for research and demonstration projects, tightening regulations, educational policies
Market formation	Factors driving new market formation, e.g. changes in customer or institutional demands and prices	Regulation inducing niche markets, fiscal policy, market-based policy instruments, public procurement, demand-side management
Legitimation	Social acceptance and compliance with institutions and society	Problem and justification framing in policies, public participation in the legislation preparation and permit processes, stimulating public debate, educational policies
Resource mobilisation	Financial and human factors and complementary assets as inputs for development	Subsidies, educational policies, secondment of expertise, provision of risk funding
Development of positive externalities	Benefits to other actors, e.g. by reducing uncertainty, cost sharing or lobbying influence	Policies for R&D funding, investment support, informational instruments, open and participatory policy processes

3. Research approach and methods

3.1. Biogas as a case

The particular biogas production systems involved in this study include co-digestion processes, where various biological wastes or by-products go through anaerobic digestion producing biogas (a mixture of mainly carbon dioxide and methane) and digestate. The materials used in the process typically include municipal biowastes, different industrial wastes, sludge from wastewater treatment plants and manure or sludge from farms. The produced biogas is a renewable fuel that can be used in energy production for heat and/or electricity, or after refining, as a traffic fuel. The digestate can be used as a fertilizer in agricultural production either directly or after different kinds of refinement processes. Thus, the potential environmental benefits of biogas systems include the replacement of fossil fuels and the reduction of emissions from landfills, for example. Biogas production also contributes to a more closed nutrient cycle, where nutrients used in crop production are recycled back to the fields after different stages of food production or consumption.

Besides the systems studied here, biogas can also be produced on a smaller scale in farms, in wastewater treatment plants as part of the wastewater treatment process, or it can be collected from landfills as landfill gas. These systems were excluded from this study, because we were interested in new connections between waste management and energy production.

In Finland, biogas production at co-digestion units is a relatively recent phenomenon. The main organic waste treatment method in Finland is composting but the role of anaerobic digestion in organic waste treatment is increasing. The first co-digestion plant was established in the early 1990s, whereas the majority of the plants have been built since 2009. Currently there are ten co-digestion plants in Finland and several new plants are under construction or at planning stage (Huttunen and Kuittinen, 2013). These plants produced 112 GWh energy in 2012 (about 0.1% of total renewable energy consumption in Finland) (Huttunen and Kuittinen, 2013; Statistics Finland, 2013). In the EU in general, biogas production is more common. One-fourth of the organic waste treatment capacity is covered by anaerobic digestion (De Baere and Mattheeuws, 2012) and 5% of renewable energy consumption is covered by biogas (Eurostat, 2012).

Biogas production connects waste management, energy production, food, and transport systems by the end-uses of the waste management process: biogas and digestate. As biogas production involves different systems it cannot efficiently be evaluated by focusing only on one system. It creates an overlapping system of its own, which blurs the boundaries of more traditional systems and, thus, has the potential to change and merge these systems.

Despite the involvement of the different systems, the core actors in Finnish biogas production come from waste management and other environmental services. Waste management companies are those making investment decisions on biogas plants for the treatment of organic waste. Only one energy company and two fuel distribution companies have been involved in building biogas plants. In addition to established waste and energy sector actors, new companies have been created around biogas production. These companies have emerged, for example, from environmental consulting and agriculture. Furthermore, one investment company has become active in creating a biogas fund and financing the building of biogas plants via one biogas company.

3.2. Data and methods

The data consisted of 20 semi-structured interviews with the core actors in the biogas field in Finland. They included co-digestion biogas plant managers (eight), representatives of companies currently planning or building a biogas plant (four) and biogas related researchers and developers (eight). All the existing biogas plants were included in the study (one did not respond to contact attempts) and other interviewees were selected using a snowball method until a saturation of perspectives was reached. The interviewees could provide first-hand knowledge on the relevant policies and their implementation effects as required in policy coherence analysis.

The interviews were conducted during spring 2013 and they addressed three broad themes: the respondent's own activities and interests in biogas sector, the kind of biogas systems suitable in Finland, and barriers and drivers in biogas systems. For the purposes of analysing policy coherence, discussions related to the third theme and direct linkages to policies were the most important. We also reviewed the relevant policies and instruments related to biogas production and that were referred to in the interviews.

We performed the analysis in two stages: (1) analysis of policy coherence from the local actor perspective, and (2) analysis of innovation effects of policy coherence. In the first stage, we focused on identifying the key points of policy coherence from the perspective of the interviewees. The policy coherence analysis focused on the three dimensions of policy coherence: internal, external and temporal. For mapping the internal and temporal coherence issues, we looked at the production system from the viewpoints of systems for (1) waste management, (2) food provision, (3) energy production and (4) transport fuels. We identified all the policies and their effects mentioned by the interviewees related to each system. After locating the policies and related coherence issues into different sectors, we identified the external effects of the policies.

In the second stage, we assessed the innovation effects of the policies and their coherence using the TIS functions for policy analysis (Table 1). System functions were compared to the policy coherence issues identified in the first stage of the analysis (Fig. 2). This was performed by three researchers separately and in case of differing opinions, the final analysis was jointly deliberated. The innovation effects were deducted based on the interviewees' accounts on the consequences of the coherence issues: how the policies and their combined effect influenced the choices made by the actors. The

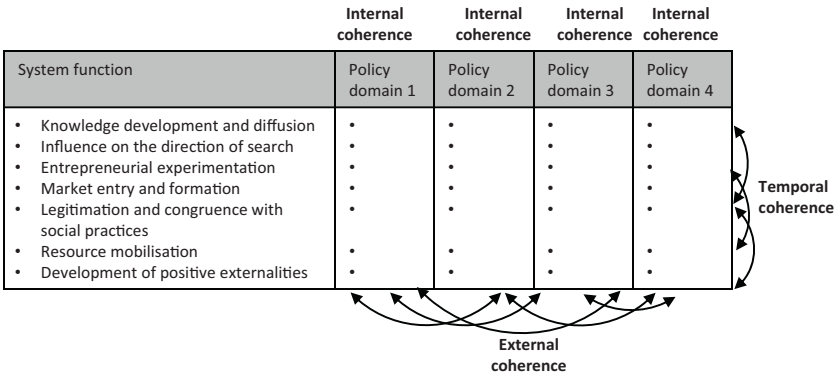

Fig. 2. Combining the analysis of innovation system functions to the analysis of policy coherence.

interviewed actors had unanimous perceptions on policy coherence and its innovation effects related to biogas production. Thus, in the analysis, no balancing between different perceptions was required (cf. Hommels et al., 2013). However, the current analysis deals only with biogas production, the policies may contain other kinds of coherence issues related to other activities.

4. Policy coherence in biogas production

Each system (energy production, food production, waste management and transport) contain policy instruments which the interviewees identified as having an influence on the biogas production system (Table 2). In addition to these direct instruments, policy strategies influence the expectations related to future options in biogas production.

The Sections 4.1–4.4 describe findings on internal and temporal policy coherence within the four systems, while the external and related temporal policy coherences are addressed in Section 4.5. The identified coherence issues are summarised in Table 3.

Table 2
Current Finnish and EU-level laws that impact on biogas production.

Policy themes	Policy instruments			
Energy	Act on Feed-in Tariff (1396/2010)	Statute on Energy Investment Subsidies (VN1063/2012)	Grid access (386/1995)	
Agriculture	Act on Fertilizer Products (539/2006) and related statute (MAF24/11)	Animal by-products Regulation (EC 1069/2009)	Measures required for agri-environmental payments and related terms	Nitrate Statute (VN931/2000)
Waste and Environment	Waste Act (646/2011) Waste Statute (331/13)	Act on Environmental Protection (86/2000)	Act on Environmental Impact Assessment (468/1994)	
Transport	Biofuels Distribution Obligation Act (1420/2010)	Act on Car Tax (5/2009) and Vehicle Tax (1327/2009)	Act on Fuel Taxes (1399/2010)	

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Table 3
Summary of the identified policy coherence issues. Coherence is marked with +, incoherence with –.

	Internal	Temporal	External
Waste management	+ Policy goals and regulations push for increasing biowaste treatment with biogas as a best option. – Waste act contains also regulations, which result in incoherent outcomes compared to policy aims. – Implementation of regulations and permit procedures (including EIA) may hamper the adoption of biogas production.	+ Temporally consistent policies push for increased biowaste treatment.	+ Coherent policy goals between waste management, food and energy production sectors + Support for R&D from all domains. – Poorly suitable fertilizer legislation clashes with supportive policies of other sectors. – Poorly functioning feed-in tariff, lack of other subsidies and limited access to grid clash with supportive policies of other sectors. – Ambiguous taxation and lack of incentives for biogas utilisation in transport (compared to liquid biofuels) clashes with general policy goals and supportive instruments of other sectors. – Poor exchange of knowledge and co-operation between officials in different sectors create incoherent implementation.
Food production	+ Policy goals push for nutrient recycling. – Regulations contain inconsistencies regarding the utilisation of digestate from biogas plants as fertilizer. – Fertilizer regulations fail to meet their goal in assuring for the safety of fertilizer products.	– Regulations related to organic fertilizers have been changing: uncertainty exists regarding future demands.	
Energy production	+ General policy goals, feed-in tariffs and investment subsidies push for biogas production. – Despite the general-level push, the feed-in tariff functions badly and access to the grid is limited.	– General policy goals have raised expectations that have not materialised as instruments in a way local actors have expected. – Sudden changes and uncertainty regarding subsidies exist	
Transport	– The fuel and vehicle taxation system creates inconsistent signals regarding biogas as fuel – Biogas fits to the general goals of carbon emission reductions and reducing oil dependency, but there are no particular aims or instruments for the promotion of biogas, such as the blending obligation for liquid biofuels		

4.1. Waste management: collection and treatment systems

Several policies have contributed to the interest of waste management companies in building a biogas plant. Both Finnish and EU-level legislation have caused a need to develop capacity related to organic waste treatment, influencing biogas companies' investment decisions. In Finland, all residential buildings with more than ten apartments are required to source-separate household-based biowaste, and the disposal of organic waste to landfills will be prohibited by 2016 (Act 646/2011 and related statute 331/2013, which implement EU's Landfill Directive (31/1999/EC) and Waste Framework Directive (2008/98/EC)).

Although the general interest in biogas production is stimulated by the policies in the waste sector, the biogas plant managers identified also problematic regulations. The Waste Act (646/2011) allows different kind of organisation basis and roles for municipalities and municipal and private companies in waste management. This liberty of organisation can result in inefficient solutions and discourage the adoption of biogas technology. For example the collection of bio-waste is not organised optimally for biogas production. Thus, the policy outcome does not respond to the policy goal due to municipalities and other waste producers implementing the Act.

A further key coherence problem identified by the biogas plant managers and by those planning a biogas plant emerges at the implementation stage related to the required environmental permit (Act 86/2000) and EIA procedures (Act 468/1994). In accordance with the EU Waste Framework directive (2008/98/EC), Finnish waste legislation is based on the principle of waste hierarchy, which means that if waste cannot be prevented or recycled it should be used first as a material, then as energy – landfill disposal being only the last option. The interviewed waste management actors regarded biogas technology as the best solution for bio-waste treatment according to the waste hierarchy. However, at the implementation level, the environmental administration was seen to prefer composting solutions and not guide municipalities towards biogas treatment in the environmental impact assessment (EIA) statements or in environmental permits.

The implementation of the EIA procedures was seen as laborious and time consuming. During the period between the first EIA announcement and the realisation of the actual environmental permit, the local situation in available biomasses and the interest of funding organisations can change considerably. These problems delay the actualisation of the plant for a few years, or may endanger the building of the plant altogether. In addition, different regional agencies differ in setting conditions for environmental permits, permit procedures and monitoring the permits, which puts actors in dissimilar positions in different parts of the country. The cross-sectoral character of regulating biogas production is clearly visible in permit-related problems, since sometimes requirements set in the environmental permit can conflict with legislation for nutrient production, for example.

4.2. Food production: nutrient recycling or how to manage the digestate?

In the food production sector, biogas production is relevant in terms of animal manure management and nutrient production. Nutrient recycling is a priority in Finnish agricultural policy, aiming to make Finland the leading country in nutrient recycling (MAF, 2011). Especially the interviewed researchers and developers, but also many of the representatives of biogas producing companies regarded biogas production as an important means to contribute to this goal.

Despite the encouraging policy goals, the same actors regarded the policy instruments steering the utilisation of the digestate to hamper the adoption of biogas technologies. This results in policy incoherence between policy goals and instruments. The utilisation of the digestate is regulated both nationally and by the EU, including the national Act on Fertilizer Products (Act 539/2006) and the related statute (MAF24/11) and the European regulation on the utilisation of animal by-products (EC 1069/2009). The regulation targets the treatment process required for different types of organic wastes and the quality of the end product (Marttinen et al., 2013). In addition, the use of fertilizers in agriculture is influenced by measures and terms related to agri-environmental payments and the Nitrate Statute (VN931/2000). Although the regulations related to fertilizer products have been developed to better account for the needs of biogas production, the interviewed biogas developers and market actors considered that the regulations are unnecessarily strict and contain inconsistencies that were

difficult to understand and accept. For example, the agri-environmental payments favour the utilisation of manure compared to the utilisation of digestate, or a fertilizer produced from digestate in a biogas plant is not accepted as a fertilizer in forestry, although it can be used in food production. The regulations related to agri-environmental payments and the Nitrate statute suffered also from temporal coherence problems as these regulations have been waiting for upcoming changes but the direction of these changes has been unclear.

The purpose of the strict fertilizer regulations is to guarantee the safety and minimise environmental harms caused by the fertilizers. Despite the regulations many farmers are still reluctant to accept digestate or other kinds of recycled fertilizers, which means that the current strict legislation has failed to reassure the relevant actors on the safety of using the digestate as a fertilizer. This indicates a coherence problem between the policy goal and the outcome, which might be caused by implementation.

4.3. Producing and using energy

In Finnish energy policy, biogas has a minor role in the overall aim of increasing the share of renewables (MEE, 2011, 2013). The major renewable energy source in Finland is forest-based biomass and also the majority of policy incentives has been focused on promoting it (see e.g. Kivimaa and Mickwitz, 2011). However, the general positive policy atmosphere around bioenergy and the fact that supporting energy from biogas has been on the government's agenda from 2007, have set high hopes for the biogas actors. Thus the interviewees were unanimously disappointed in the feed-in tariff for electricity produced using biogas (Act 1396/2010), because was actualised as late as in 2011, and it included conditions (e.g. utilisation of new components, laborious and expensive reporting procedures) that made the tariff unavailable or unattractive for many biogas producers. Furthermore, both the market actors and biogas developers criticised the feed-in-tariff for focusing only on electricity production and not facilitating other energy uses of biogas such as injection to gas grid or direct utilisation in industrial processes. Thus the Feed-in-tariff Act demonstrated temporal incoherence due to the time lag between the goal setting and the policy instrument and internal incoherence between the goal and the requirements in the instrument. By 2013 only one biogas producer had applied to the feed-in tariff system.

After the introduction of the feed-in tariff, a tax-subsidy for renewable electricity was abolished. This meant that electricity produced from biogas is taxed similarly to electricity produced from fossil fuels. This decision was rather sudden, showing elements of temporal policy incoherence, and, for the biogas actors, it increased the feeling of uncertainty related to energy policy. From the perspective of companies planning for new biogas plants, the investment subsidies in place were also considered uncertain, since the subsidy level and types of eligible energy production are subject to changes, another case of temporal policy incoherence.

Regulations related to grid access have proven to be problematic for biogas plants (Act 386/1995) demonstrating incoherence between the Act and the goal of guaranteeing grid access. Electricity production at biogas plants occurs at a relatively small scale and although the access to the grid is guaranteed, each producer has to negotiate with the grid operator on the price and terms of the grid access. Transfer costs related to the available price for the electricity are high. The biogas developers and company representatives regarded the energy governance as being designed for industrial scale actors, whereas small and medium scale distributed production lacks proper incentives and fits poorly into the general system.

4.4. Transport

The transport sector was regarded as the most difficult from policy perspective. Most of the biogas producers had considered the production of transport fuels as an option, but the current lack of policy incentives combined to high expenses makes it unattractive. In the transport sector, the utilisation of biogas is influenced by the problem of creating demand, when there is no fuel supply and vice versa. According to the Transport Policy Report to Parliament, the main aims of transport policy in Finland include carbon emission reductions and reducing the dependence on oil (MTC, 2012). However, this is

mainly done via increasing the share of liquid biofuels; Finland has introduced a biofuels distribution obligation (Act 1420/2010), which means that biofuels are blended with fossil petroleum. Promoting the use of biogas has not been central in transport policy and as pointed out by a three researchers and developers, the utilisation of biogas in transport suffers from some inconsistencies in the legislation. For example, the carbon emission-based car tax (Act 5/2009) and vehicle tax (Act 1327/2009) contain problems, which result in situations where cars that are able to use methane are taxed more highly than cars using gasoline. However, biogas is exempt from fuel taxes (Act 1399/2010). Thus, transportation related taxation combined to the ambiguous role of biogas in transport policy goals create incoherent signals to biogas actors demonstrating both vertical incoherence between policy goals and instruments and horizontal incoherence in the policy instruments.

4.5. External policy coherence

The identified internal and temporal coherence issues have external implications. The internal incoherences are multiplied as external incoherence in a situation where the policy goals (except for transport policies) clearly pushed for increased biogas production (Table 3). For example, the problems in nutrient recycling reduce the attractiveness of electricity production from biogas. Similarly, supportive policy goals in one sector contribute positively across sectors. For example, the energy production potential of biogas production, general policy goals for increasing bioenergy production and the promise of feed-in tariffs were important for triggering interest in biogas in waste management.

The analysis of external policy coherence from the actor perspective raised two policy coherence issues which were unrelated to issues identified in internal policy coherence analysis. First, poor exchange of knowledge and co-operation existed across sectors resulting in external incoherence in implementation. For example, environmental permits can conflict with nutrient legislation due to environmental officials not being aware of all the relevant legislation. This forced the actors planning biogas production to function as information mediators between the different government sectors. This has led to local problem-solving and increased the problem of regional differences in implementation, particularly with respect to environmental permits. Second, interviewees from different backgrounds reported innovation policy instruments supporting the development of biogas technologies, particularly in the form of support for biogas production related research projects. The research projects typically involved a particular problem, such as developing the fertilising use of the digestate or feasibility studies on the transport utilisation of biogas in the region. By contributing to a specific problem, they also contributed to the development of biogas in other domains. On the other hand, for the companies, research funding was not easy to access and significant R&D funding of one's own was also required.

5. Innovation effects of identified policy coherence issues based on TIS functions

The innovation effects of the policy coherence issues identified in the second stage of the analysis were fitted into the system functions (Table 4). Some coherence problems related to implementation were difficult to fit into the system function classification. These included the environmental permit procedures, and problems related to grid access. These were fitted by including market entry to 'market formation'. In the case of market formation, the inclusion of market entry clarifies the situation, although similar matters are involved in mere market formation (Jacobsson and Bergek, 2011). The system function of positive externalities was omitted from the innovation effect examination, because the external policy coherence issues identify externalities by definition. Positive external coherence issues related to innovations create positive externalities and negative create negative externalities.

The positive and negative effects of the spreading of biogas and related innovating mainly accumulated on different system functions related to all sectors. This means that some system functions were fortified by policy coherence/incoherence, while some functions were weakened. The actor-based analysis carried out in this study identified very few policies linked to 'knowledge development and diffusion'.

The positive innovation effects came mainly from the supportive externally coherent policy goals between waste management, food and energy production sectors. These generated interest in the

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Table 4
Identified innovation effects of internal, temporal and external policy coherence from local actors' perspective in a framework for TIS for policy analysis. A positive innovation-triggering effect is marked with (+) and a negative with (−).

System function		Waste management using AD	Nutrient recycling from biogas plant	Biogas for energy	Biogas for transport fuel
Knowledge development and diffusion	*Internal*				
	Temporal				
	External	Support for R&D on biogas from all policy domains (+), Poor exchange of knowledge and co-operation between officials in different sectors create incoherent implementation (−)			
Influence on the direction of search	*Internal*	Supportive policy goals and regulations have raised interest in anaerobic digestion as best option (+)	Supportive policy goals, but regulations related to organic fertilizer discourage (+/−)	Supportive policy goals for bioenergy in general, supportive feed-in-tariff, which malfunctions in practice (+/−)	While fuels taxation supports biofuels, the car and vehicle tax create inconsistent signals regarding biogas as fuel (+/−)
	Temporal	Temporally consistent policies push for increased biowaste treatment (+)	Changing fertilizer regulations cause uncertainty regarding future demands (−)	Changing subsidy policies (−)	
	External	Coherent policy goals between waste management, food and energy production sectors (+), Coherently supportive instruments between domains (+), Internally unsupportive instruments create incoherent signals with supportive regulations and goals in other domains (−)			
Entrepreneurial experimentation	*Internal*	Policy goals and regulations push for increasing biowaste treatment (+), Implementation of permit procedures is incoherent with policy goals (−)	Regulations contain inconsistencies regarding the utilisation of digestate from biogas plants as fertilizers (+/−)	Poor instruments for better electricity price has pushed for the development of alternative energy utilisation forms (+), General policy goals have raised expectations that have not materialised as instruments discouraging development (−)	
	Temporal			Changing subsidy policies (−)	
	External	Poor instruments for better electricity price has pushed for the development of traffic fuel projects (+), Coherently supportive instruments between domains (+), Internally unsupportive instruments create incoherence with supportive regulations and goals in other domains (−)			

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Table 4 (*Continued*)

System function		Waste management using AD	Nutrient recycling from biogas plant	Biogas for energy	Biogas for transport fuel
Market entry and formation	*Internal*	Regulations push for increasing biowaste treatment (+), Waste act regulations allow a diversity of ways to organise collection hampering biowaste supply for biogas plants (−), Implementation of permit procedures hinder market entry (+/−)	Fertilizer regulations hamper market formation and entry for the fertilizer produced from the digestate (−)	Despite the general-level push, the feed-in tariff functions badly and access to the grid is limited (−),	While fuels taxation supports biofuels, the car and vehicle tax create inconsistent signals regarding biogas as fuel (+/−)
	Temporal		Uncertain changes in fertilizer regulations hamper market entry (−)	Changing subsidy policies hamper market entry (−)	
	External	Internally unsupportive instruments create incoherent signals between regulations in other domains (−)			
Legitimation	*Internal*	Positive policy goals and instruments have increased the legitimacy of biogas technologies (+)	Positive policy goals have increased the legitimacy, but fertilizer regulations fail to meet their goal in assuring for the safety of fertilizer products (+/−)	General policy goals and feed-in tariff push for biogas and increase legitimacy (+)	Ambiguous policies may decrease legitimacy (−)
	Temporal				
	External	Positive policy goals and instruments have increased legitimation (+), Incoherent instruments have decreased legitimation (−)			
Resource mobilisation	*Internal*	Implementation of permit procedures complicate the availability of funding (−)	Regulations related to organic fertilizers include subsidies for the utilisation of manure as a fertilizer, which discourages biogas technologies (−)		Blending obligation directs resources for the development of liquid biofuels instead of biogas (−)
	Temporal			Changing subsidy policies (−)	
	External	Internally unsupportive instruments create incoherence between regulations and goals in other domains (−)			

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'search for biogas systems', 'entrepreneurial experimentation' and 'legitimation'. This effect was fortified by external policy coherence related to available R&D funding pushing for 'knowledge development and diffusion'. The positive policy goals were supported by regulations for waste management, and to some extent also by the feed-in tariff, which further enhanced these functions via both internal and external coherence.

In 'entrepreneurial experimentation', these innovation-triggering effects included internal and external policy *incoherence*, where the lack of policy instruments or their poor applicability and unsupportiveness have actually resulted in innovating: In the food sector, fertilizer regulations created internal and external policy incoherence. Yet simultaneously, the regulations posed requirements that needed to be addressed, creating a demand for innovations, which has pushed for 'entrepreneurial experimentation' in some sense. Thus, strict and perceivably unsuitable legislation has also been an innovation trigger for different digestate refining technologies, and also for process optimisation. Similarly in the energy sector, the lack of supporting instruments for the utilisation of biogas in electricity production has forced biogas producers to consider options broadly in order to find the most suitable combination for energy production (including the consideration of traffic fuel projects). Often, at least part of the produced biogas is utilised in nearby industrial facilities. This way, adequate value for the product can be obtained. These arrangements are innovative and often resource-efficient.

Minor and less straightforward positive innovation effects were produced by the internal and temporal coherence in waste management regulations, which created a market pull for the application of new biowaste treatment solutions facilitating 'market entry and formation'. Furthermore the waste management regulations accelerated competition for different biowaste materials and this triggered companies to develop their services to be able to accept specific valuable organic waste types suitable for biogas production. A typical example is packaged food from grocery stores, where plastic packages need to be separated from the organic material. However at the municipal level, the implementation of waste regulations clashed with the policy aims, downplaying the positive effect on 'market entry and formation'.

The negative innovation effects followed mainly from regulations failing to meet the policy goals and the resulting internal and external policy incoherence. Also the confusing transportation policies discouraged innovation. The negative effects were most clearly observable in 'market entry and formation' and 'resource mobilisation' related to especially agricultural, energy and transport sectors. Also, temporally changing energy and agricultural policies downplayed innovation effects related to 'direction of search', 'entrepreneurial experimentation', 'market entry and formation' and 'resource mobilisation'.

6. Discussion

The number and role of different policy domains linked to biogas systems has country-specific variation. For example, in Denmark and the Netherlands, the policy focus on biogas technologies has shifted from energy production to manure management and the reduction of nutrient emissions (Lybaek et al., 2012; Negro et al., 2007). These policy changes have created an unstable environment that does not take into account all the system aspects of biogas production and use. Similarly, in Finland, biogas systems are regarded from the perspective of one policy domain at a time, missing the overall picture of biogas production. Our actor-based policy coherence analysis pointed out the interlinkages between relevant policies influencing the expansion and diffusion of biogas systems, and further innovating related to them in Finland. These interlinkages influence the actions taken by actors and, therefore, also potential innovating actions.

The actors interviewed provided a consistent picture of the key problems related to different policies influencing biogas production in co-digestion plants. This was probably due to biogas production being a relatively narrow case and the fact that all the interviewed actors wanted to facilitate biogas production.

Specific points of concern regarding the incoherence of policies were mismatches between general policy goals and instruments (internal and external incoherence), temporally shifting energy policies (temporal incoherence) and the complex intertwinement of poorly functioning or implemented instruments in the different policy domains (internal and external incoherence). The

instrument-related problems are mainly caused by instruments targeting domain-specific parts of the biogas production system with inadequate attention on instruments regulating other domains. This has resulted in regulating the same actors in an inconsistent manner (cf. Flanagan et al., 2011). In turn, consistencies and synergies were observed in the general policy goals between policy domains (external coherence). The observed results support the findings of Nilsson et al. (2012), who also found policies to often be coherent in their objectives but incoherent at the level of instruments and implementation.

In a top-down analysis it would have been difficult to identify all the relevant policies and specific problems. For example, problems involving grid access, the usability of the feed-in tariff for biogas, and, particularly, the relevance of instruments steering fertilizer products and the terms and measures related to agri-environmental subsidies would have been difficult to identify. These problems have particular impacts on the diffusion of niche innovations as they hamper the market entry and market formation for biogas innovations.

The ineffectiveness of implementing policy goals into practice is one form of policy incoherence, often missed out in top-down analyses. Our results point out that the synergies existing between policy domains at the level of biogas-related policy objectives are not realised at the instrument level, as only one policy domain has sufficient instruments in place to support biogas production, others suffering from internal incoherence between goals and instruments. For example, problems related to the utilisation of the digestate reduce the attractiveness of producing biogas for energy. Resolving the problems related to the utilisation of digestate could improve the diffusion of biogas systems by easing the pressure of making energy production more profitable, for example. Thus, a positive cycle could be created. Positive cycles can also be created via specific policy instruments, since one policy instrument may act as an innovation trigger related to multiple system functions and across policy domains. R&D support and investment subsidies are good examples as the developments made in one part of the biogas system have external influences to other parts as well. Similarly, malfunctioning regulation and fiscal instruments multiply their unsupportive effect across TIS functions and even policy domains as the fertilizer regulations show.

If innovations in biogas production are to be promoted by policies, specific attention in all sectors should be paid to promoting market entry and formation, knowledge development and diffusion, and resource mobilisation. More generally, the policy coherence approach to the system functions points to the importance of holistic assessment of all the system functions and policies influencing them. Policies simultaneously influence several system functions and care should be applied when planning instruments targeted at the amelioration of a specific function (cf. Hekkert et al., 2007).

From an innovation-triggering perspective, incoherent policies seem to have a dual nature. In some cases, policy incoherence or conflict can be seen as a specific innovation-triggering force, for example, when market pull created by waste policies clashes with agricultural, transport and, partially also, energy policies, inducing the actors to find solutions to survive in the situation. This is comparable to situations where tightening environmental regulation triggers innovation (van den Bergh, 2013; Kemp and Pontoglio, 2011). On the other hand, policy incoherence discourages the diffusion of innovations, the biogas systems. It can also be questioned whether the innovations triggered by policy incoherence are guiding innovations in a desirable direction.

We found that the system functions help in dividing the innovation effects of policy coherence issues further into target areas and point out which system functions are relevant in terms of policy coherence – in this empirical context at least, influence on the direction of search, entrepreneurial experimentation and market formation. The actor perspective used in the analysis was beneficial in identifying the relevant policies for the different system functions. Compared to Kivimaa and Virkamäki (2014), several policies related to legitimisation could also be identified. Further empirical studies would give insights into the significance of particular functions for policy coherence in more general sense.

7. Conclusions

The policy coherence and actor perspectives on policy analysis were successful in identifying critical points for policy development and how they affect biogas innovation diffusion. The Technological

Innovation System (TIS) functions helped in identifying critical points in terms of policy coherence. By combining the TIS functions and policy coherence analysis, we have provided new theoretical insights into how policy coherence and different system functions are related in innovation processes. Most importantly, the role of actors in both the analysis of policy coherence and innovations should be developed further.

The analysis of biogas production demonstrates how policies related to one domain also have supporting effects in meeting goals of other policy domains. This policy push clashes with unsupportive policy instruments in many domains. Yet, in some cases, it may result in actors finding innovative solutions. Incoherent policies can, therefore, act as a source of innovation. However, from the perspective of biogas production as a common biowaste treatment method, the coherence of policies would more likely facilitate the diffusion. Moreover, with respect to innovation support for biogas production in Finland, discouraging policy effects existed in 'market entry and formation' and 'resource mobilisation'. Related to 'knowledge development', there is a policy gap.

In the Finnish biogas case, the existing coherent and plausible general policy goals should be reflected at the instrument level. Therefore, careful mapping of the policy instrument needs should be made in all four policy domains related to biogas production. Furthermore, grassroots actors, especially the boundary-crossing ones, should be involved in policymaking processes as they are able to inform the policymakers on the practical realities of biogas production. Finally, in designing policies for innovation, it is vital to account for the regime boundary-crossing policy coherence, bearing in mind that incoherence also has innovation-triggering potential.

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